Digestion and Nutrient Absorption

Glossary of Terms

- **Duodenum** – The first section of the small intestine. It is .3 meters long.
- **Jejunum** – The second section of the small intestine. It is 1-2 meters long.
- **Ileum** – The third section of the small intestine. It is 1.5 meters long.
- **Hepatic Portal Vein** – The vein that delivers nutrients rich blood to the liver.
- **Esophageal Sphincter** – A gateway separating the esophagus and the stomach, controlling the entry of food.

Glossary of Terms

- **Pyloric sphincter** – A gateway separating the stomach and the small intestine, controlling the entry of chyme.
- **Chyme** – A slushy acidic mixture of food and digestive juices.
- **Enzymes** – Protein molecules that increase the speed of chemical reactions in the body. They work by combining with and altering the molecules of other chemical substances. The digestive enzymes split large molecules of food into smaller units for absorption.

Glossary of Terms

- **Villi** – Projections along the small intestinal wall increasing area surface for greater absorption.
- **Hydrolysis** – splitting of substances by adding water
- **Glycerol** – the combination of fats and oils.

The gastrointestinal tract:

- Includes the esophagus, gall bladder, liver, stomach, pancreas, small intestine, large intestine, rectum, and anus (which are both part of the large intestine).
- Its entire length from the mouth to the anus is 6-8 meters long.
- It is the tube that supplies nutrients and water to the body.

The GI Odyssey

- The nutrients are delivered to the liver by way of the hepatic-portal vein and then they are distributed throughout the body by the circulatory system.
- It takes the GI tract 1-3 days to eliminate the ingested food.

Mouth & Esophagus

- By chewing, grinding, mashing and softening food particles, food becomes easier to swallow and becomes accessible to enzymes and other digestive substances that start the breakdown process.
- Smooth muscle contracts rhythmically causing the contents to move (peristalsis).
- This happens at zero gravity even if a person is turned upside down.
- At the end of the esophagus is the esophageal sphincter which is a one way ring of muscle that relaxes to allow the food to enter the stomach.

The Stomach

- The stomach's volume can range from .5 liters when empty to up to 6 liters when distended after an extremely large meal.
- Inside the stomach wall are gastric glands. These glands secrete hydrochloric acid and other powerful enzyme containing digestive juices that continually degrade the nutrients.

The Stomach

- Carbohydrates are the easiest macronutrient to digest and leave the stomach most rapidly, followed by proteins than fats.
- It takes between 1-4 hours to empty depending on the nutrient concentration and volume of the meal.
- The food mixes with a chemical substance called chyme, and then passes through the pyloric sphincter into the small intestine

The Small Intestine

- Approximately 90% of digestion (and essentially all lipid digestion) takes place in the first two sections of the 3 meter long small intestine.
- It is a coiled structure with three sections called the duodenum, jejunum and the ileum.
- There are tiny fingerlike protrusions along the walls of the small intestine called villi.
- These structures absorb the carbohydrates, proteins, lipids, water, vitamins and minerals.

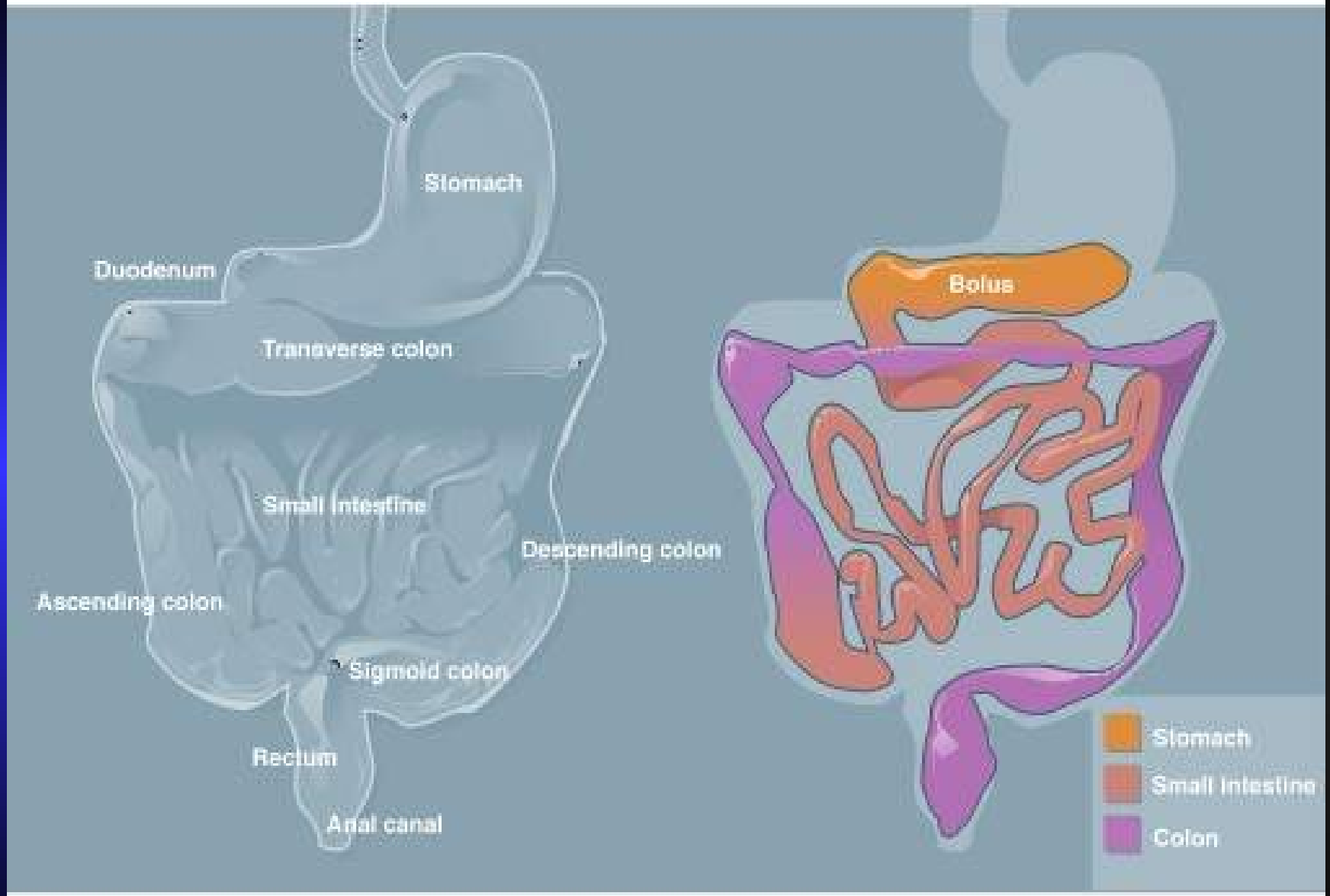
The Small Intestine

- The intestinal walls are made up of smooth muscle that contract and relax moving the food or chyme forward then slightly backward.
  - This gives the intestine additional time for absorption.
  - Here the pancreas secretes 1.2-1.6 liters of alkali containing juice to help buffer the hydrochloric acid that mixes with the chyme and enters into the small intestine from the stomach.
- Neutralizing this acid is crucial otherwise it causes ulcerations or ulcers.

The Large Intestine

- This is the final digestive structure.
- It consists of the ascending and descending colon, transverse colon, sigmoid colon, rectum and anal canal. It is 1.2 meters long.
- By the time the digested food or chyme reaches the large intestine, most of the nutrients have been absorbed.
- The primary role of the large intestine is to convert chyme into feces for excretion.
- Here the colon absorbs water from the chyme, changing it from liquid to solid. The large intestine does not contain villi.

3.13. Gross anatomical structures of the large intestine.



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The Digestive Process

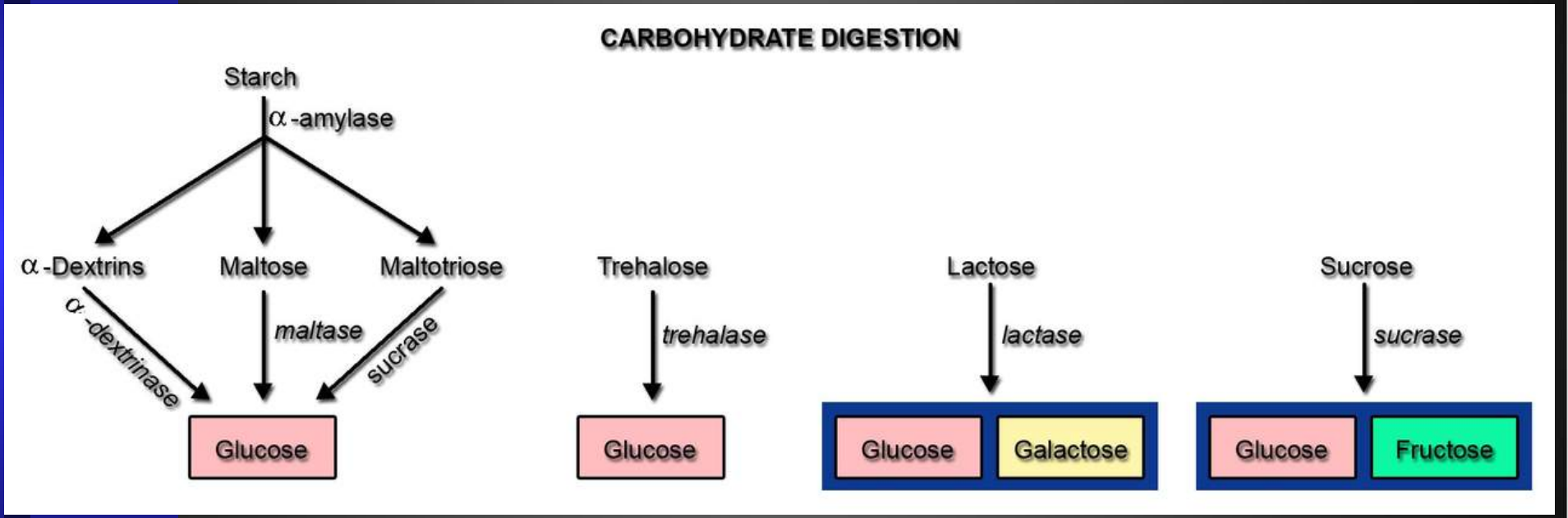
Carbohydrate digestion and absorption

- Carbohydrates start breaking down in the mouth.
- The enzyme salivary amylase starts breaking up the starches reducing it into smaller glucose molecules where in the stomach it continues to be broken down further.
- Upon entering the small intestine the pancreas releases the enzyme pancreatic amylase to help complete the hydrolysis of starch into smaller chains of glucose molecules – monosaccharides, which is 1 molecule of sugar.

The Digestive Process

(Carbohydrate digestion continued)

- The monosaccharides are absorbed into the small intestine and delivered to the liver by way of the hepatic portal vein.
- After the liver processes the nutrients, the nutrients enter into the blood stream circulating throughout the body.


CARBOHYDRATE DIGESTION
Starch
α-amylase
α-Dextrins
Maltose
Maltotriose
α-dextrinase
maltase
sucrase
Glucose
Trehalose
trehalase
Glucose
Lactose
lactase
Glucose
Galactose
Sucrose
sucrase
Glucose
Fructose

The Digestive Process

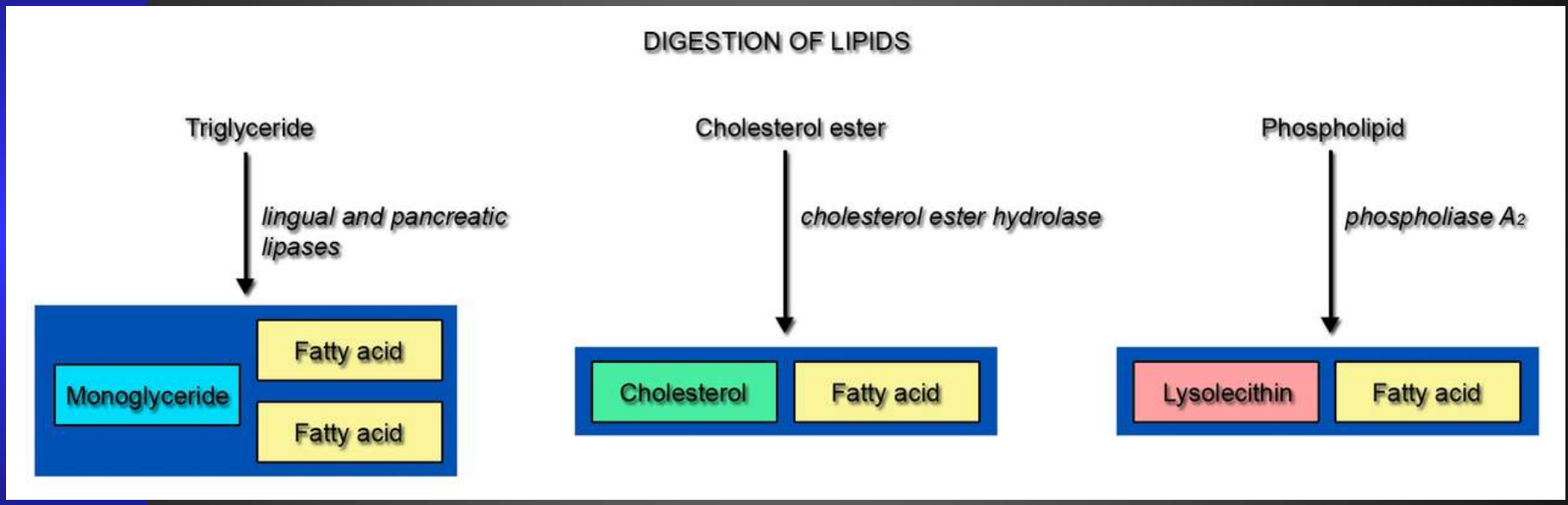
Lipid or fat digestion and absorption

- In the stomach fats are broken down into glycerol and fatty acids.
- The digestion of fats starts in the stomach when mixed with the enzyme lipase.
- The major part of the breakdown takes place in the small intestine.
- In the duodenum the enzyme pancreatic lipase furthers the process by breaking the fats down from triglycerides to monoglycerides (which is 1 fatty acid instead of 3 fatty acids connected to a glycerol molecule).

The Digestive Process

(Lipid/fat digestion and absorption continued)

- Bile is produce in the liver and secreted by the gall bladder which increases the lipids solubility, breaking it down into droplets, making it easier for the small intestine to absorb.
- When foods with high lipid content enter the stomach, the hormone – gastric inhibitory peptide is released, slowing down movement flow out of the stomach.
- This is why we feel full after eating high fat foods.


DIGESTION OF LIPIDS
Triglyceride
lingual and pancreatic lipases
Monoglyceride
Fatty acid
Fatty acid
Cholesterol ester
cholesterol ester hydrolase
Cholesterol
Fatty acid
Phospholipid
phospholiase A2
Lysolecithin
Fatty acid

The Digestive Process

Protein Digestion and Absorption

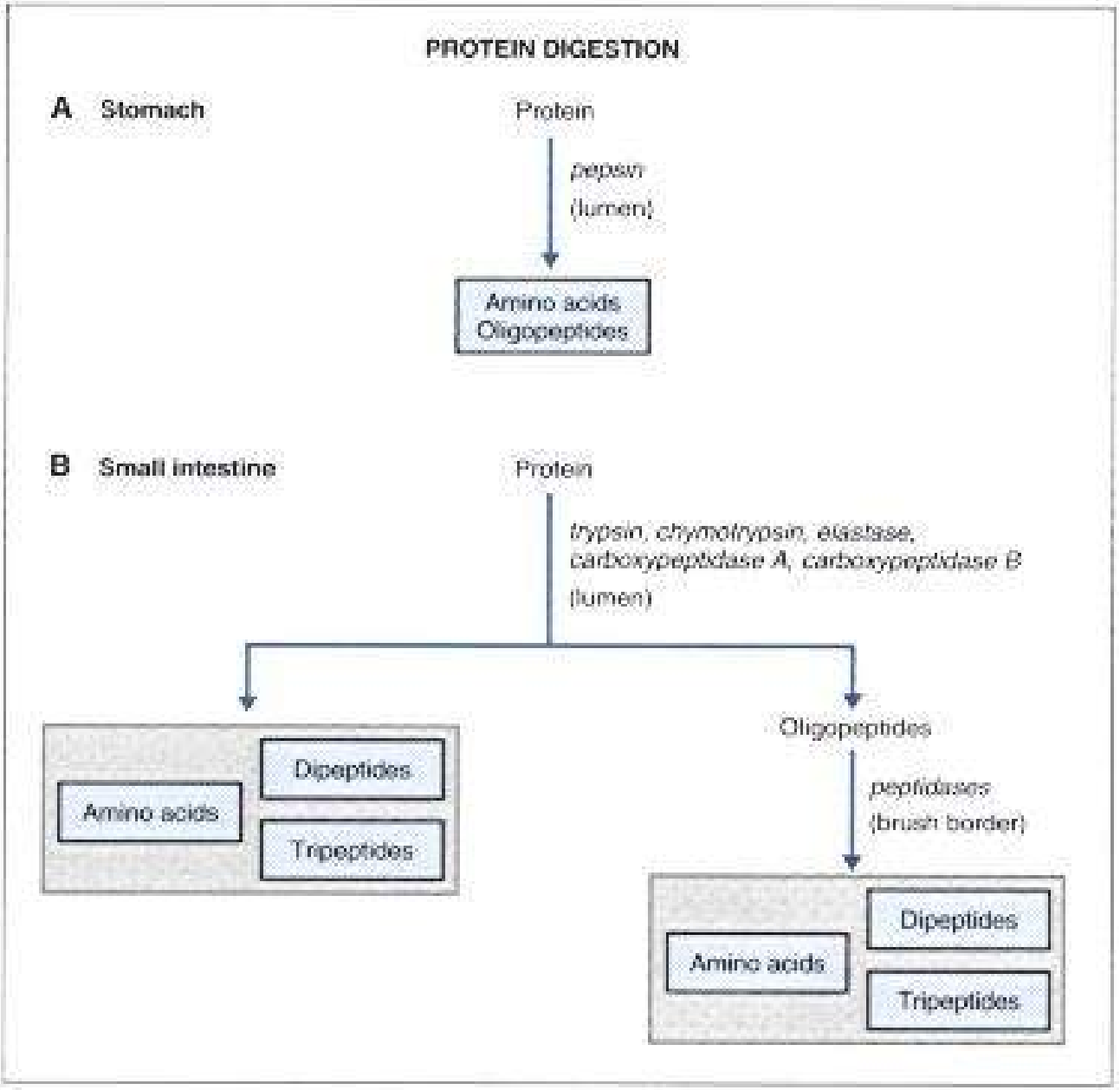
- Proteins are split into linked amino acids called peptides and then into individual amino acids.
- In the stomach the enzyme pepsin starts the breakdown of proteins into smaller units called polypeptides and peptides.
- In the duodenum of the small intestine the pancreatic enzymes trypsin and chymotyrpsin also split proteins into polypeptides and peptides.

The Digestive Process

(Protein Digestion and Absorption continued)

- In the jejunum of the small intestine an enzyme created by the small intestine called peptidase splits the large peptides into smaller peptides and than into amino acids.

- All of these smaller protein fragments go directly to the liver by the hepatic portal vein.
- Once in the liver one of three things happens to the proteins:
  - 1. It converts to glucose,
  - 2. It converts to fat or
  - 3. It is directly released into the blood as amino acids.


PROTEIN DIGESTION
A Stomach
Protein
pepsin
(lumen)
Amino acids
Oligopeptides
B Small intestine
Protein
trypsin, chymotrypsin, elastase,
carboxypeptidase A, carboxypeptidase B
(lumen)
Amino acids
Dipeptides
Tripeptides
Oligopeptides
peptidases
(brush border)
Amino acids
Dipeptides
Tripeptides

The Digestive Process

Vitamin Absorption

- Vitamins are absorbed mainly in the jejunum and ileum sections of the small intestine. A, D, E, and K are the fat soluble vitamins and they must be absorbed in combination with fat.
- Fat is the transport for which the vitamins are delivered to the liver.

The Digestive Process

(Vitamin Absorption Continued)

- The B's and C vitamins are water soluble vitamins.
- They start breaking down in the stomach and then throughout the small intestine.
- They do not remain in the body's tissues very long and the excess is passes on into the urine.

The Digestive Process

Mineral Absorption

- Intestinal absorption of minerals increases when the body is lacking the particular nutrient.
- Intestinal absorption of minerals decreases when the body is not lacking the particular nutrient.

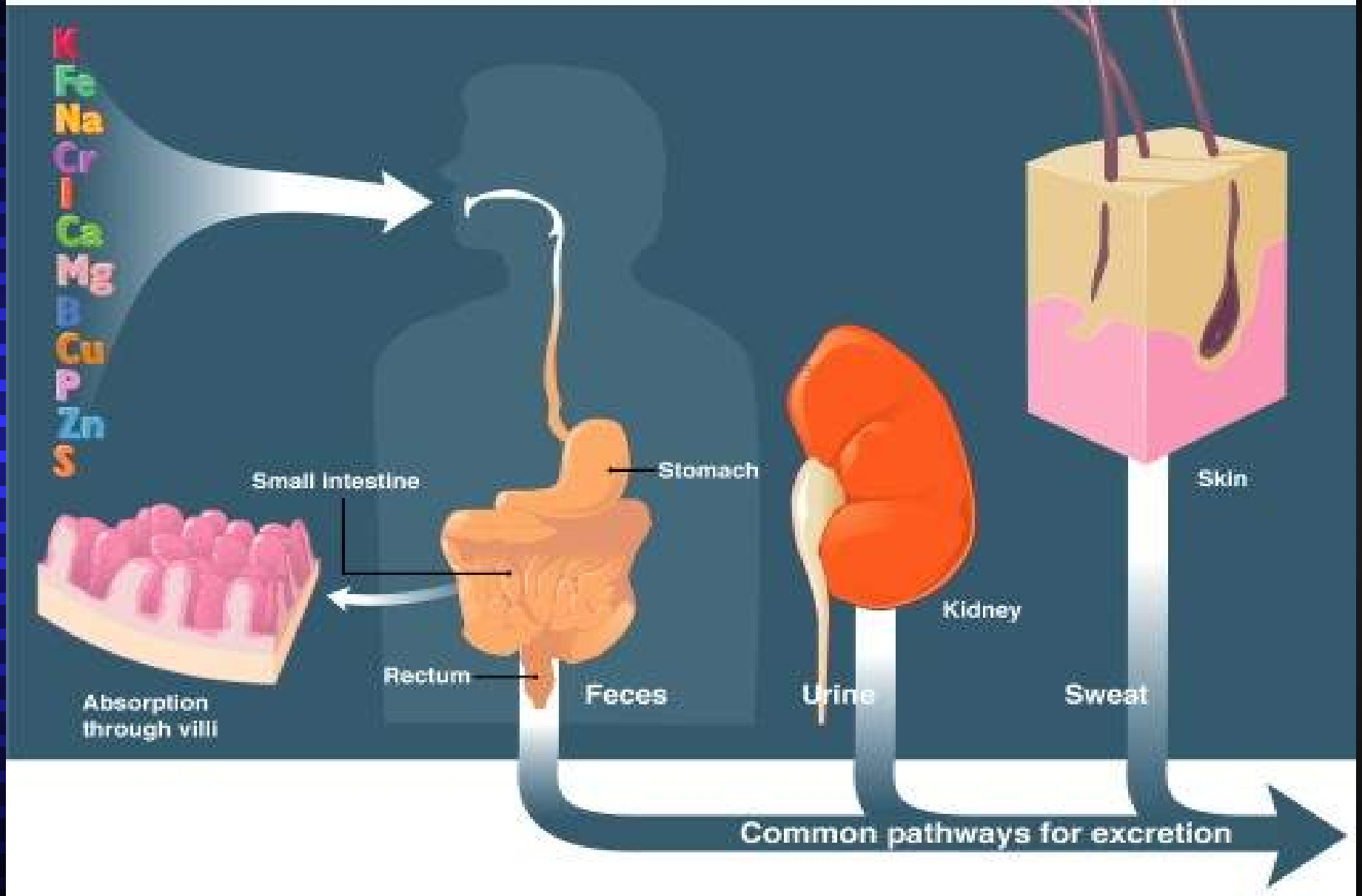
The Digestive Process

Water Absorption

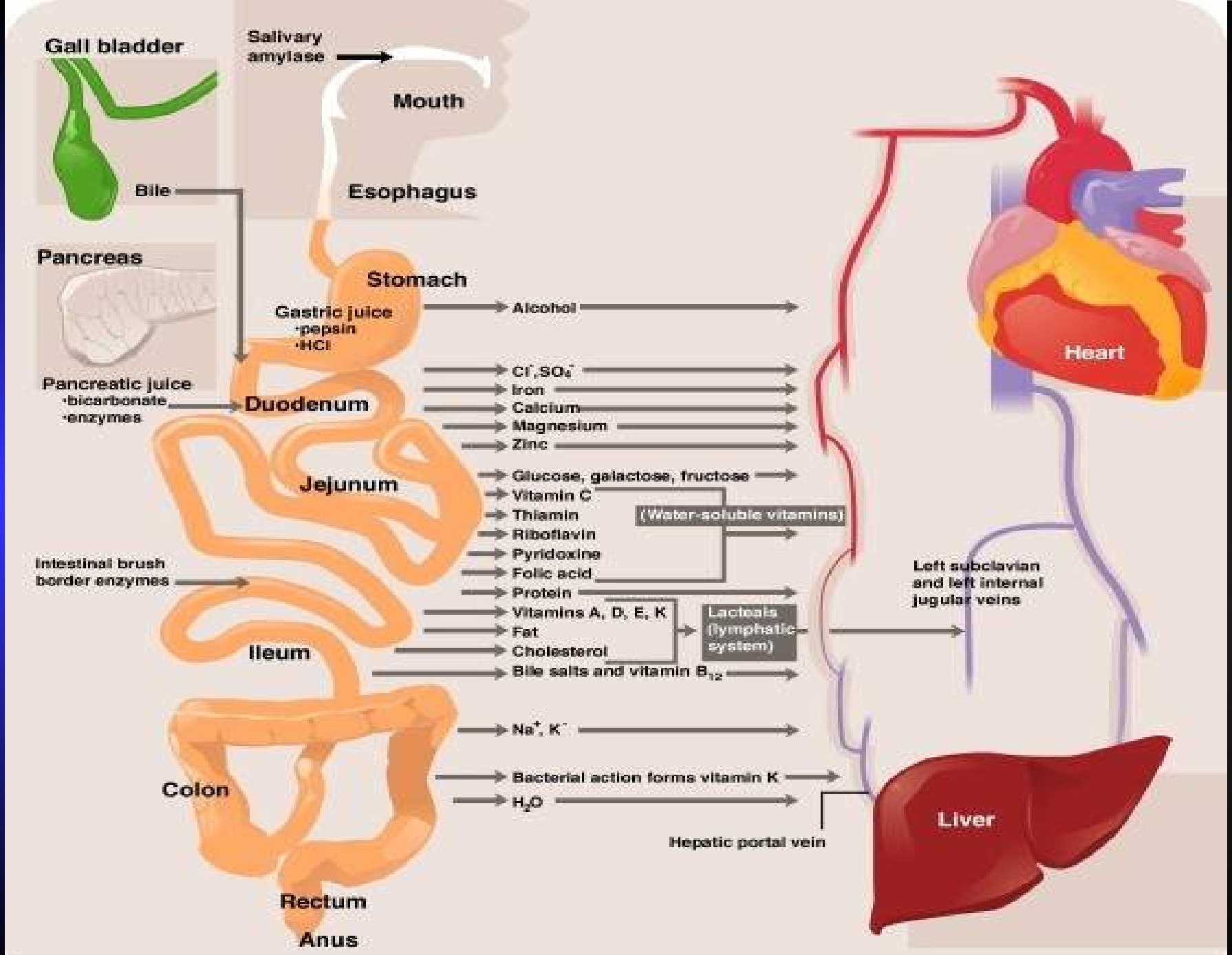
- Saliva, gastric secretions, bile, pancreatic acids and intestinal secretions take up approximately 7 liters of water, when combined with an intake of an average of 2 liters of water ingested, the body is processing approximately 9 liters of water daily.

- 72% is absorbed in the first half of the small intestine, 20% is absorbed in the lower half of the small intestine, and 6% is absorbed in the large intestine.

3.17. Absorption of minerals and common pathways of excretion.

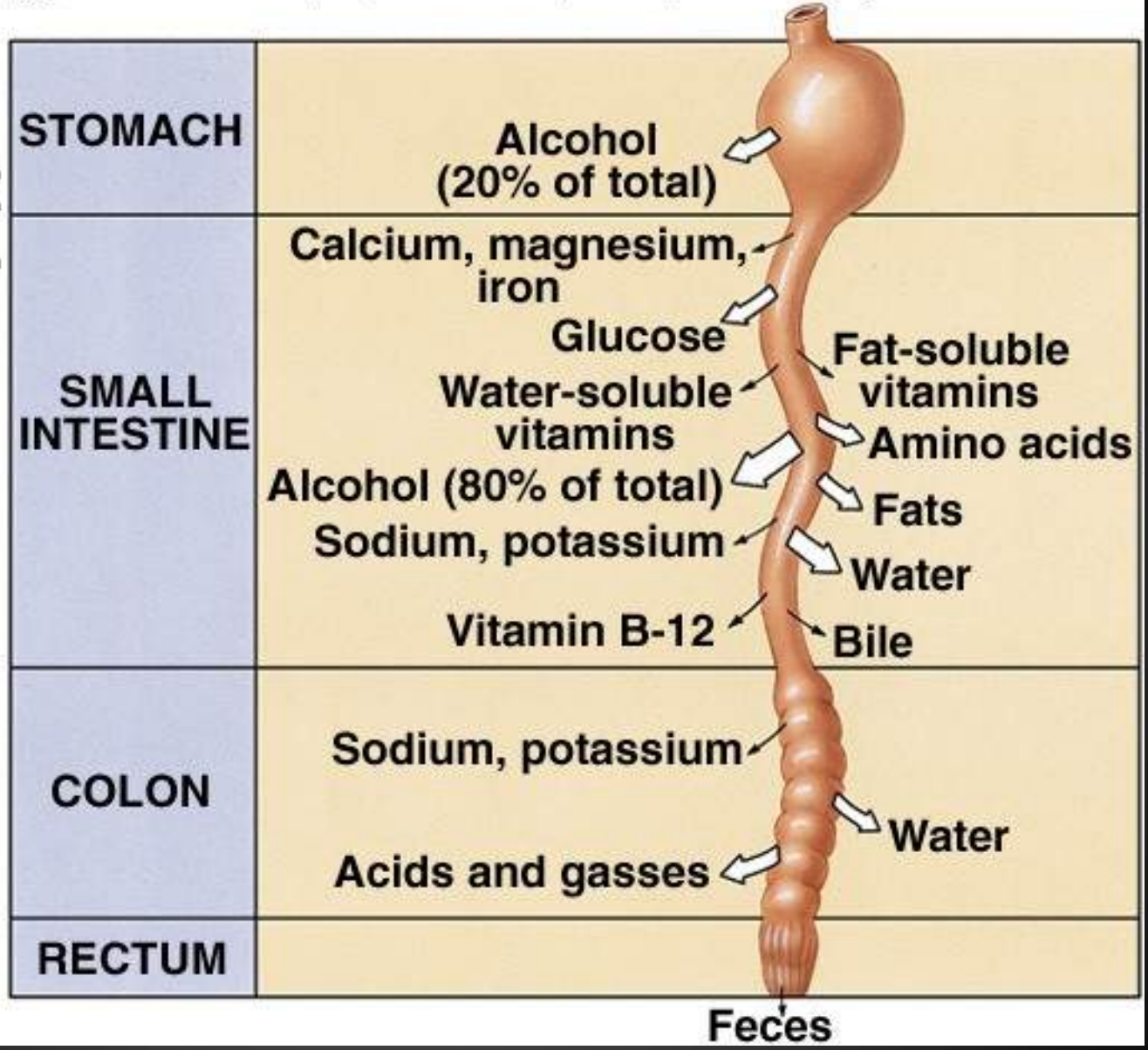


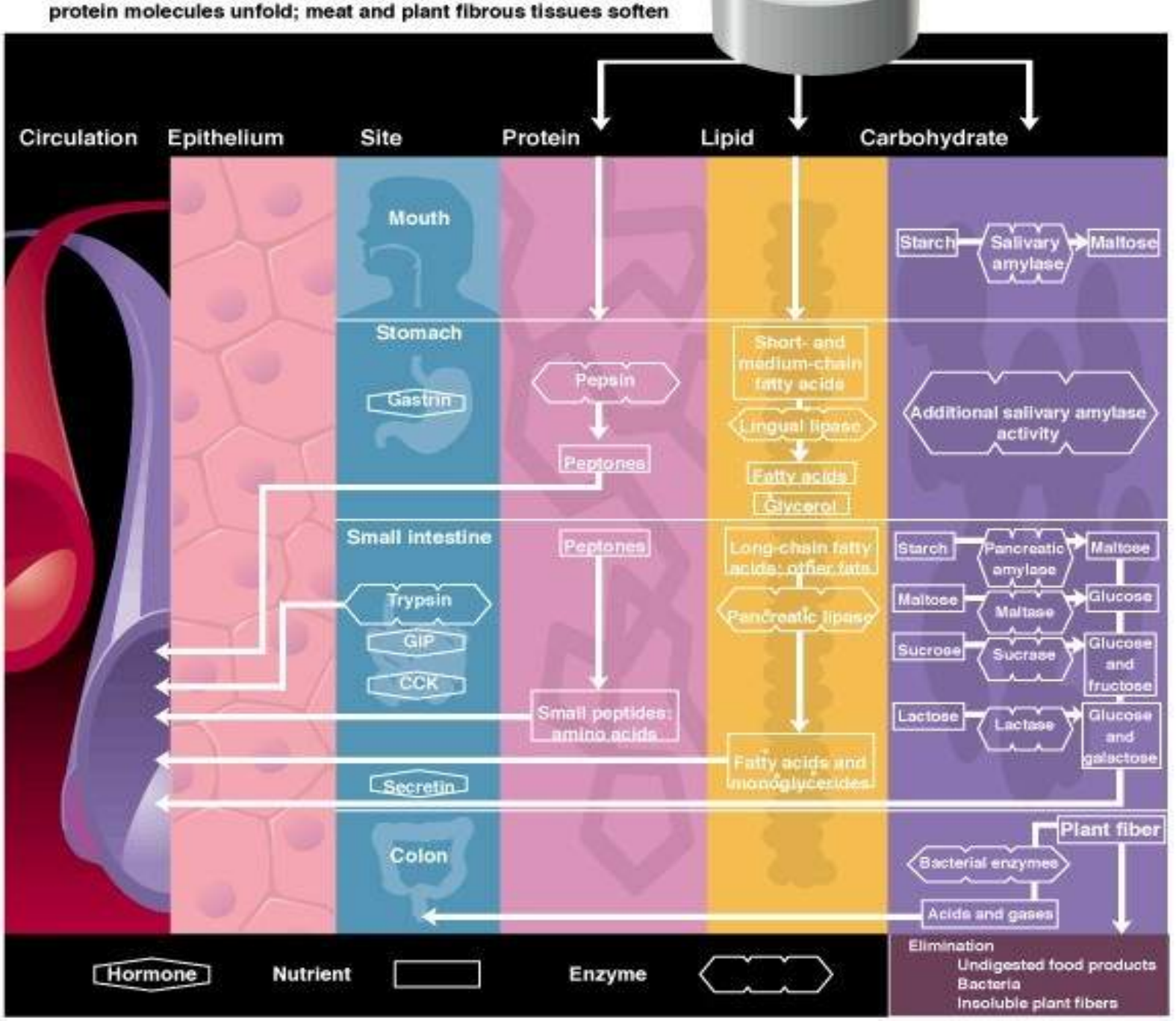
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Gall bladder
Salivary amylase
Mouth
Bile
Esophagus
Pancreas
Stomach
Gastric juice
·pepsin
·HCl
Alcohol
Pancreatic juice
·bicarbonate
·enzymes
Duodenum
Cl^-, SO_4^-
Iron
Calcium
Magnesium
Zinc
Jejunum
Glucose, galactose, fructose
Vitamin C
Thiamin
Riboflavin
Pyridoxine
Folic acid
(Water-soluble vitamins)
Intestinal brush border enzymes
Protein
Vitamins A, D, E, K
Fat
Cholesterol
Lacteals (lymphatic system)
Ileum
Bile salts and vitamin B_{12}
Left subclavian and left internal jugular veins
Heart
Na^+, K^-
Colon
Bacterial action forms vitamin K
H_2O
Liver
Hepatic portal vein
Rectum
Anus

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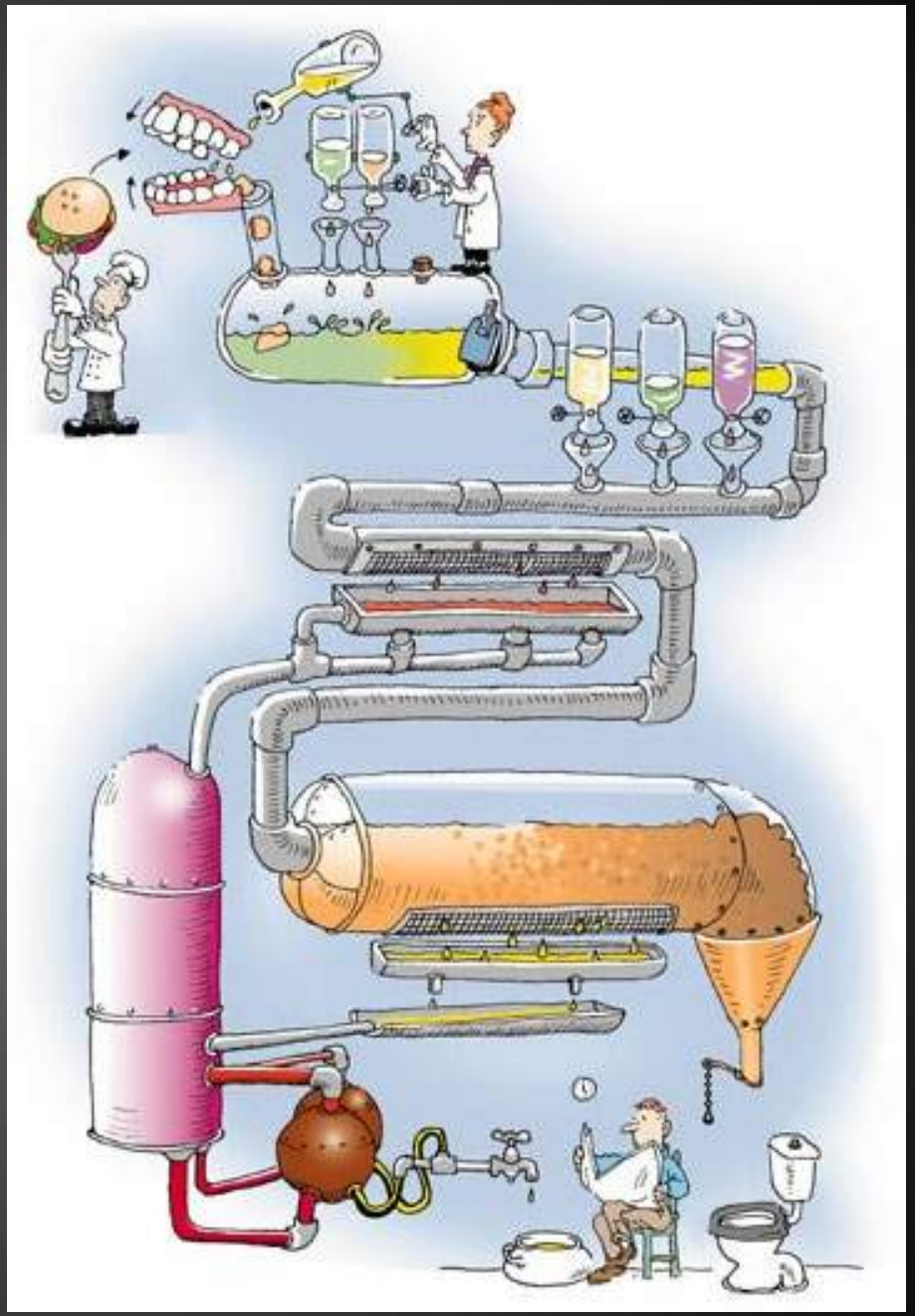
Sites of Nutrient Absorp-tion





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- good digestion and absorption of nutrients demand good health: sleep, activity, state of mind, as well as diet





Thank You!!!